

OpenSOC SEA-NREN Initiative

Empowering Research Networks Against
Cyber Threats

Building Cybersecurity Resilience for Southeast Asia's
Education and Research Networks



Executive Summary

Emerging National Research and Education Networks (NRENs) across Southeast Asia face growing cybersecurity threats, yet many lack the resources, expertise, and infrastructure to defend themselves effectively. Without dedicated Security Operation Centers (SOCs), these networks remain vulnerable to attacks that can disrupt research, education, and innovation.

The OpenSOC SEA-NREN Initiative addresses this challenge by enabling NRENs to deploy open-source SOCs, supported by structured training and regional collaboration. This initiative fosters a proactive cybersecurity culture, ensuring that institutions can detect, respond to, and mitigate threats in real time.

Key Objectives & Benefits

Our initiative focuses on **four** strategic pillars:

- **Capacity Building:** Equip NREN personnel with practical skills through structured training and hands-on SOC deployment.
- **Infrastructure Development:** Deploy open-source SOCs tailored for resource-constrained environments, ensuring cost-effectiveness and scalability.
- **Regional Collaboration:** Establish a threat intelligence-sharing network to enable collective defense and knowledge exchange.
- **Policy & Governance:** Support NRENs in developing cybersecurity policies that institutionalize best practices and ensure sustainability.

Why Partner With Us

Partnering with us means contributing to a safer, more resilient digital ecosystem for education and research across Southeast Asia. This initiative aligns with APNIC Foundation's strategic outcomes and UN Sustainable Development Goals, including Industry, Innovation & Infrastructure, Quality Education, and Partnerships for the Goals.

BitRanger brings a unique combination of technical expertise and regional insight. As a licensed SOC provider approved by Malaysia's National Cyber Security Agency (NACSA) and a spin-off from the Cybersecurity Research Centre (CYRES) at Universiti Sains Malaysia, BitRanger has successfully delivered SOC solutions and capacity-building programs. Our track record includes international projects under AsiaConnect and SecureAsia, demonstrating our ability to execute complex, multi-country initiatives with measurable impact.

Impact & Sustainability

This project is designed for long-term impact. By combining skills development, infrastructure deployment, and collaborative threat intelligence, we create a foundation for sustained cybersecurity resilience. Our approach ensures that participating NREs can maintain and expand SOC capabilities beyond the project lifecycle, supported by open resources, community engagement, and ongoing academic collaboration.

Call to Action

Join us in building a resilient cybersecurity ecosystem for Southeast Asia's education and research networks. We invite you to collaborate, participate, or support this transformative initiative.

Contact: +60124062348 (Whatsapp)

Email: opensoc@bitranger.io

Website: www.bitranger.io